Python for Supply Chain

Course outline

Course title	Python for Supply Chain
Class dates	From 16/11/2026 to 17/12/2026
Format	Online, 10 sessions, 15:00 SGT (14:00 WIB), 90-minute slot

1. Overview

Excel is great, until a file is too big, the steps repeat every week, or you need to show exactly what you did. Python helps with all three. In this course you start from zero and work with real sales data from a UK online gift shop. You clean the data, find which products matter, show it in clear charts, and present one finding in 5 minutes.

2. Course objectives

1. Read and fix common Python errors.
2. Load, check and clean a real CSV file with pandas.
3. Summarise data with groupby and sort products with ABC analysis.
4. Make clear charts with matplotlib and seaborn.
5. Present a finding in 5 minutes, answer first.

3. Assessment

Component	Weight
Class participation and daily recall	20%
Cold test	15%
Project	30%
Final presentation	20%
Self-reflection	15%
Total	100%

This course is not graded. The weights show what matters most.

4. Schedule and outline

Lesson	Date	Session	Resources	Deadlines
1	Mon 16 Nov	Hello, Python. Print, variables, types, round, f-strings, reading errors.	Notebook, handout, slides	
2	Wed 18 Nov	Shelves, cards and choices. Lists, dictionaries, comparisons, if / elif / else.	Notebook, handout, slides	
3	Mon 23 Nov	Let the loop do the walking. For loops, functions, reading a traceback.	Notebook, handout, slides	Mount Google Drive once
4	Thu 26 Nov	Cleaning day. Load a real CSV, check it, clean it, add columns. (May use 5 spare minutes.)	Notebook, handout, slides, dataset credit	
5	Mon 30 Nov	Who are the star products? Groupby, running total, ABC analysis.	Notebook, handout, slides	Cold test (in class)
6	Thu 3 Dec	Make the numbers talk. Merge and charts with matplotlib and seaborn.	Notebook, handout, slides, case hunt template	Case hunt handed out
7	Mon 7 Dec	New tools, your own question. A taste of SQL and Power BI, then your project plan.	Notebook, handout, slides, sql-practice.com, templates	Project plan
8	Thu 10 Dec	Detective work. Case hunt talk, then project work.	Notebook, handout, slides	Case hunt due
9	Mon 14 Dec	Polish and publish. Final charts, three sentences, GitHub, slides.	Notebook, handout, slides, templates	Project and GitHub; reflection handed out
10	Thu 17 Dec	Showtime. Your 5-minute presentation, questions, feedback, look back.	Slides, scoring sheet	Reflection (before the session); presentation

5. General guide and resources

Dataset: Chen, D. (2012). Online Retail II [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5CG6D. Licence: CC BY 4.0.

Course home: one page with the schedule and your folders in Google Drive. Open it first.

Notebooks: open each notebook from Google Drive in Google Colab. Run the cells from the top, in order.

Daily recall: three short questions a day between sessions. Answer from memory first, then check.

Help: write down what is unclear at the end of each session and bring it to the next one.

6. Academic honesty

Your work must show your own understanding. Credit every source and the dataset.